

Appendix C

LINEAR PROGRAMMING

Inequality is the cause of all local movements.
—Leonardo da Vinci

Chapter Outline

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C.1 INTRODUCTION

Linear programming is perhaps *the* core model of constrained optimization; and the simplex method for solving linear programming has been one of the most significant algorithmic discoveries of this century. Developed in 1947, the simplex method has stood the test of time, having been applied to thousands of applications in fields as diverse as agriculture, communications, computer science, engineering design, finance, industrial and military logistics, manufacturing, transportation, and urban planning. Moreover, methods and concepts developed for linear programming—such as duality theory, decomposition methods, and sensitivity analysis—have served as important base methodologies for stimulating discoveries in many other fields within the sphere of optimization. For these reasons, linear programming rightly deserves its position as one of the basic cornerstones of applied mathematics, computer science, and operations research.

In this appendix we summarize some of the basic ideas of linear programming and the simplex method. We do so for at least two reasons. First, a great majority of the models that we have developed in this book, and indeed most of the models encountered in the field of network optimization, are either linear programs or integer programming extensions of linear programs. Therefore, a firm understanding of linear programming is valuable for understanding the structure of the models we have been studying. Second, although we have attempted to develop much of network flows from first principles and to use basic combinatorial ideas rather than more general methodologies and concepts of linear programming, we have, by necessity, needed to invoke ideas from linear programming on many occasions, sometimes as a basic tool in our development and at other times to make appropriate connections between the ideas we have been developing and more general concepts. Therefore,

we have needed to rely on several central ideas of linear programming. This appendix serves to make our coverage as complete as possible; it functions both (1) as an introduction to linear programming for those who have only passing familiarity with this topic, and (2) as a review of linear programming for those readers who are already conversant with this topic.

A linear program is an optimization problem with a linear objective function, a set of linear constraints, and a set of nonnegativity restrictions imposed upon the underlying decision variables; that is, it is an optimization model of the form

$$\text{Minimize } \sum_{j=1}^q c_j x_j \quad (\text{C.1a})$$

subject to

$$\sum_{j=1}^q a_{ij} x_j = b(i) \quad \text{for all } i = 1, \dots, p, \quad (\text{C.1b})$$

$$x_j \geq 0 \quad \text{for all } j = 1, \dots, q. \quad (\text{C.1c})$$

This problem has q nonnegative decision variables x_j and p equality constraints (C.1b). (In many texts, m denotes the number of equality constraints and n denotes the number of decision variables. This notation, unfortunately, is the reverse of the convention in network flows, since network flow systems contain one constraint per node and one variable per arc. For this reason we do not use the notation of m and n to denote the number of constraints and variables of a linear program.)

We assume, by multiplying the i th equation by -1 , if necessary, that the right-hand-side coefficient $b(i)$ of each constraint $i = 1, \dots, p$ is nonnegative. We might note that we could formulate a linear program in several alternative ways; for example, the objective function could be stated in maximization form, or the constraints could be in a less than or equal to or greater than or equal to form. The linear programming literature frequently refers to the formulation (C.1)—a model with equality constraints, nonnegative variables, and a minimization form of the objective function—as the *standard form* of a linear program.

In economic planning, each decision variable models one particular production activity (x_j is the level of that activity), each constraint corresponds to a scarce resource, and the coefficient a_{ij} indicates the amount of the i th resource consumed per unit of the j th production activity. In this instance the model seeks the “best” use of the scarce resources, that is, the production plan that uses the available resources to produce the maximum possible revenue (assuming a maximization form of the objective function).

In matrix notation, the linear programming model has the following form:

$$\text{Minimize } cx \quad (\text{C.2a})$$

subject to

$$Ax = b, \quad (\text{C.2b})$$

$$x \geq 0. \quad (\text{C.2c})$$

In this formulation the matrix $A = (a_{ij})$ has p rows and q columns, the vector $c = (c_j)$ is a q -dimensional row vector, the vectors $x = (x_j)$ and $b = (b(i))$ are q -

and p -dimensional column vectors, respectively. We let $\mathcal{A}_j$ denote the column of $\mathcal{A}$ corresponding to the variable x_j . We assume that the rows of the matrix are linearly independent; thus the system $\mathcal{A}x = b$ contains no redundant equations. In terms of linear and matrix algebra (we assume modest background concerning these topics), this assumption states that the rows of the matrix $\mathcal{A}$ are linearly independent; that is, the matrix $\mathcal{A}$ has full row rank.

For the special case of the minimum cost network flow problem, each component of the decision variable x corresponds to the flow on an arc and the matrix $\mathcal{A}$ has one row for each node of the underlying network. In this case the matrix $\mathcal{A}$ is the node-arc incident matrix $\mathcal{N}$ that we introduced in Chapter 1.

In the following sections we describe the rudiments of linear programming theory. We begin by illustrating the underlying geometry of linear programs and by introducing the fundamental concepts of extreme points and basic feasible solutions. Then we describe the key features of the simplex method and of a variant that permits us to efficiently handle upper bounds on the decision variables. We conclude this appendix by introducing the basic features and key results of linear programming duality theory.

C.2 GRAPHICAL SOLUTION PROCEDURE

Linear programs involving only two or three variables have a convenient graphical representation that helps in understanding the nature of linear programming and of the simplex method. We illustrate this procedure using the following example, stated in inequality form with a maximization objective:

$$\text{Maximize } z = x_1 + x_2 \quad (\text{C.3a})$$

subject to

$$2x_1 + 3x_2 \leq 12, \quad (\text{C.3b})$$

$$x_1 \leq 4, \quad (\text{C.3c})$$

$$x_2 \leq 3, \quad (\text{C.3d})$$

$$x_1, x_2 \geq 0. \quad (\text{C.3e})$$

The shaded region in Figure C.1 is the set of feasible solutions for this problem. The set of feasible solutions for the linear programming problem is generally referred to as a *polyhedron*. The points A , B , C , D , and E are the *extreme points* of the polyhedron; these points are formed by the intersection of the lines corresponding to various constraints. Note that extreme points do not lie on any line segment joining two other points in the polyhedron. More formally, a vector x is a strict convex combination of distinct vectors $x^1, x^2, \dots, x^k$ if $x = \theta^1 x^1 + \theta^2 x^2 + \dots + \theta^k x^k$ for a set of weights $\theta^i > 0$ satisfying the condition $\sum_{i=1}^k \theta^i = 1$. A vector x is an extreme point of a polyhedron if it is not a strict convex combination of two distinct points in the polyhedron, that is, cannot be represented as $x = \theta x^1 + (1 - \theta)x^2$ for some weight $0 < \theta < 1$ and two distinct points x^1 and x^2 of the polyhedron. It is an easy exercise to show that if x is a strict convex combination of $k > 2$ distinct points in the polyhedron, it is not an extreme point.

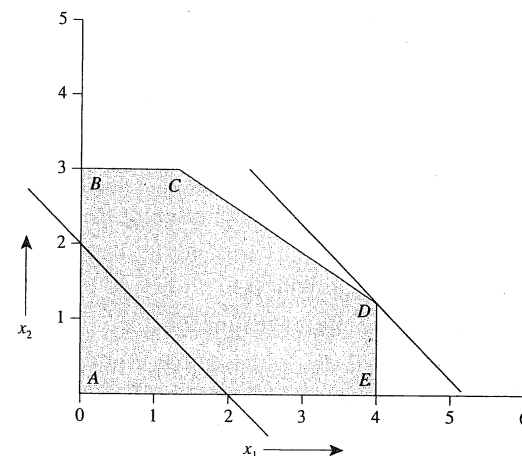


Figure C.1 Set of feasible solutions for a linear program.

The linear programming problem seeks a point (x_1, x_2) in the polyhedron $ABCDEA$ that achieves the maximum possible value of $x_1 + x_2$. Equivalently, we wish to determine the largest value of w for which the line $x_1 + x_2 = z$ has at least one point in common with the polyhedron $ABCDEA$. The lines obtained for different values of z are parallel to each other. Since these lines move farther away from the origin as z becomes larger, to maximize $x_1 + x_2$ we need to slide the line $x_1 + x_2 = z$ away from the origin as far as possible so that it has some point in common with the polyhedron $ABCDEA$. We can do so until we reach some extreme point, point D in our case, at which point the line has only one point in common with the polyhedron; for any further translation of the line away from the origin, no matter how small, the line has no point in common with the polyhedron. Thus point D is an optimal solution of our linear programming problem; its objective function value equals $16/3$.

This graphical solution procedure illustrates an important property of linear programs, namely that every linear program always has an extreme point solution as one of its optimal solutions. Therefore, to solve a linear programming problem, we can focus only on the extreme point solutions. Consequently, we need to consider only a finite number of solutions. The simplex method, described in the next section, makes use of this extreme point property. It starts at some feasible extreme point and visits "adjacent" extreme points, improving the objective function values of the solution at each step, until it reaches an optimal extreme point. For instance, in our example, if the simplex method starts at point A , it might visit the points B and C before reaching the optimal extreme point D . Alternatively, it might follow the path A, E, D .

C.3 BASIC FEASIBLE SOLUTIONS

Our description of the simplex method requires that the linear program to be solved be stated in the standard form (C.1); that is, the objective function is in the minimization form; each constraint, except for the nonnegative condition imposed on

the decision variables, is an equality; and each right-hand-side coefficient $b(i)$ is nonnegative. Any linear program not in the standard form can be brought into standard form through simple transformations. For example, maximizing $\sum_{j=1}^q c_j x_j$ is equivalent to minimizing $-\sum_{j=1}^q c_j x_j$. To model an inequality constraint $\sum_{j=1}^q a_{ij} x_j \leq b(i)$ as an equality constraint, we could add a new nonnegative "slack variable" y_i , with zero cost, and writing the inequality as $\sum_{j=1}^q a_{ij} x_j + y_i = b(i)$.

During its execution, the simplex method modifies the original linear program stated in the standard form by performing a series of one or more of the following elementary row operations:

1. Multiplying a row (i.e., constraint) by a constant, or
2. Adding one row to another row or to the objective function.

Since we have stated all the constraints in the equality form, row operations do not affect the set of feasible solutions of the linear program. As an illustration, consider the following linear program:

$$\text{Minimize } z(x) = x_1 + x_2 - 8x_3 + 6x_4 \quad (\text{C.4a})$$

subject to

$$2x_1 + x_2 - 14x_3 + 10x_4 = 16, \quad (\text{C.4b})$$

$$x_1 + x_2 - 11x_3 + 7x_4 = 10, \quad (\text{C.4c})$$

$$x_1, x_2, x_3, x_4 \geq 0. \quad (\text{C.4d})$$

Subtracting (C.4c) from (C.4a) and (C.4b) gives the following equivalent linear program:

$$\text{Minimize } z(x) = 0x_1 + 0x_2 + 3x_3 - x_4 + 10 \quad (\text{C.5a})$$

subject to

$$x_1 - 3x_3 + 3x_4 = 6, \quad (\text{C.5b})$$

$$x_1 + x_2 - 11x_3 + 7x_4 = 10, \quad (\text{C.5c})$$

$$x_1, x_2, x_3, x_4 \geq 0. \quad (\text{C.5d})$$

Subtracting (C.5b) from (C.5c) gives another equivalent formulation of this linear program:

$$\text{Minimize } z(x) = 0x_1 + 0x_2 - 3x_3 + x_4 + 10 \quad (\text{C.6a})$$

subject to

$$x_1 - 3x_3 + 3x_4 = 6, \quad (\text{C.6b})$$

$$x_2 - 8x_3 + 4x_4 = 4, \quad (\text{C.6c})$$

$$x_1, x_2, x_3, x_4 \geq 0. \quad (\text{C.6d})$$

Since the linear program (C.6) is equivalent to (C.4), both have the same optimal solutions. A linear program like (C.6) is said to be in the *canonical form* if it satisfies the following *canonical property*.

Canonical Property. The formulation has one decision variable isolated in each constraint; the variable isolated in a given constraint has a coefficient of +1 in that constraint and does not appear in any other constraint, nor does it appear in the objective function.

The previous formulation satisfies the canonical property because x_1 is isolated in constraint (C.6b) and x_2 in constraint (C.6c).

A linear program typically has a large number of canonical forms since there are many ways to isolate decision variables in the constraints. The canonical form in (C.6) has the following attractive feature. Assigning any values to x_3 and x_4 uniquely determines the values of x_1 and x_2 . In fact, setting $x_3 = x_4 = 0$ immediately gives the solution $x_1 = 6$ and $x_2 = 4$. Solutions such as these, known as *basic feasible solutions*, play a central role in the simplex method. In general, given a canonical form for any linear program, we obtain a basic solution by setting the variable isolated in constraint i , called the i th *basic variable*, equal to the right-hand side of the i th constraint, and by setting the remaining variables, called *nonbasic*, all to value zero. Collectively, the basic variables are known as a *basis*.

In general, we obtain a basic feasible solution as follows. We isolate a variable in each constraint. For simplicity, assume that we have isolated the variable x_i in the i th constraint. Let $\mathbf{B} = \{1, 2, \dots, p\}$ denote the index set of basic variables and let $\mathbf{L} = \{p+1, p+2, \dots, q\}$ denote the index set of nonbasic variables (we choose the mnemonic $\mathbf{L}$ for the index set of nonbasic variables because we set the value of the corresponding variables to their lower bound—that is, value 0—in the basic solution defined by this index set). We refer to the pair $(\mathbf{B}, \mathbf{L})$ as a *basis structure* of the linear problem. For a given basis structure $(\mathbf{B}, \mathbf{L})$, we can compatibly partition the columns of the constraint matrix $\mathcal{A}$. Let $\mathcal{B} = [\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_p]$ and $\mathcal{L} = [\mathcal{A}_{p+1}, \mathcal{A}_{p+2}, \dots, \mathcal{A}_q]$. We refer to the $p \times p$ matrix $\mathcal{B}$ as a *basis matrix*. We also let $\mathbf{x}_{\mathbf{B}} = [x_i : i \in \mathbf{B}]$ and $\mathbf{x}_{\mathbf{L}} = [x_j : j \in \mathbf{L}]$ be a partitioning of the variables into subvectors corresponding to the index sets $\mathbf{B}$ and $\mathbf{L}$. With this notation, we can rewrite the constraint matrix $\mathcal{A}\mathbf{x} = \mathbf{b}$ as

$$\mathcal{B}\mathbf{x}_{\mathbf{B}} + \mathcal{L}\mathbf{x}_{\mathbf{L}} = \mathbf{b}. \quad (\text{C.7})$$

We can convert (C.7) to the canonical form by premultiplying each term by $\mathcal{B}^{-1}$, the inverse of the basis matrix, giving

$$\mathbf{x}_{\mathbf{B}} + \mathcal{B}^{-1}\mathcal{L}\mathbf{x}_{\mathbf{L}} = \mathcal{B}^{-1}\mathbf{b}. \quad (\text{C.8})$$

We obtain a *basic solution* from (C.8) by setting each nonbasic variable to value zero. The resulting solution is

$$\mathbf{x}_{\mathbf{B}} = \mathcal{B}^{-1}\mathbf{b} \quad \text{and} \quad \mathbf{x}_{\mathbf{L}} = \mathbf{0}.$$

We refer to this solution as a *basic feasible solution* if the value of each basic variable is nonnegative (i.e., $\mathbf{x}_{\mathbf{B}} \geq \mathbf{0}$). We also say that the basis structure $(\mathbf{B}, \mathbf{L})$ is *feasible* if its associated basic solution is feasible. For some choices of the basis matrix, the corresponding basic solution will be feasible, and for some other choices it will not be feasible.

Converting a linear program to a canonical form requires that we invert the basis matrix, which is possible only if the columns associated with the basic variables

are linearly independent. If the associated columns are linearly dependent, the basis matrix is singular (i.e., its determinant is zero) and we cannot invert it. We shall therefore henceforth refer to a basis as a subset of p variables whose corresponding columns are linearly independent.

During its execution, the simplex method requires information about $\mathcal{B}^{-1}\mathcal{A}_j$, the updated column corresponding to the nonbasic variable x_j . We let $\bar{\mathcal{A}}_j = \mathcal{B}^{-1}\mathcal{A}_j$ and call this vector the *representation* of $\mathcal{A}_j$ with respect to the basis matrix $\mathcal{B}^{-1}$ (or, alternatively, basis $\mathbf{B}$). We refer to this column vector as a representation because by premultiplying both sides of the equations $\bar{\mathcal{A}}_j = \mathcal{B}^{-1}\mathcal{A}_j$ by $\mathcal{B}$, we obtain $\mathcal{B}\bar{\mathcal{A}}_j = \mathcal{A}_j$, which implies that we can interpret the elements in the vector $\bar{\mathcal{A}}_j$ as weights that we use to multiply the columns of the basis matrix $\mathcal{B}$ in order to obtain the column $\mathcal{A}_j$. For notational convenience, we let $\bar{\mathcal{A}}_{\mathbf{L}}$ denote the matrix $\mathcal{B}^{-1}\mathcal{L}$ containing the column representations of all the nonbasic variables and let $\bar{b} = \mathcal{B}^{-1}b$; we refer to the vector $\bar{b}$ as the modified right-hand side. Finally, let $c_{\mathbf{B}} = (c_1, c_2, \dots, c_p)$ and $c_{\mathbf{L}} = (c_{p+1}, c_{p+2}, \dots, c_q)$ denote the cost vectors associated with the basic and nonbasic variables, respectively.

In a canonical form of a linear program, each basic variable has a zero coefficient in the objective function. We can obtain this special form of the objective function by performing a sequence of elementary row operations (i.e., multiplying constraints by some multipliers and subtracting them from the objective function). Any sequence of elementary row operations is equivalent to the following: Multiply each constraint i by a number $\pi(i)$ and subtract it from the objective function. This operation gives the equivalent objective function $\sum_{j=1}^q c_j x_j - \sum_{i=1}^p \pi(i) [\sum_{j=1}^q a_{ij} x_j - b(i)]$, or, collecting terms and letting $z_0 = \sum_{i=1}^p \pi(i)b(i)$,

$$z(x) = \sum_{j=1}^p [c_j - \sum_{i=1}^p \pi(i)a_{ij}]x_j + \sum_{j=p+1}^q [c_j - \sum_{i=1}^p \pi(i)a_{ij}]x_j + z_0. \quad (\text{C.9})$$

To obtain a canonical form, we select the vector π so that

$$c_j - \sum_{i=1}^p \pi(i)a_{ij} = 0 \quad \text{for each } j \in \mathbf{B}. \quad (\text{C.10})$$

In matrix notation we select π so that

$$\pi\mathcal{B} = c_{\mathbf{B}}.$$

In this expression, $c_{\mathbf{B}} = (c_1, c_2, \dots, c_p)$ is the cost vector associated with the basic variables. We refer to

$$\pi = c_{\mathbf{B}}\mathcal{B}^{-1}$$

as the *simplex multipliers* associated with the basis $\mathbf{B}$ and refer to $c_j^{\pi} = c_j - \sum_{i=1}^p \pi(i)a_{ij}$ as the *reduced cost* of the variable x_j . Note that $z_0 = \pi b = c_{\mathbf{B}}\mathcal{B}^{-1}b = c_{\mathbf{B}}x_{\mathbf{B}}$, which since $(x_{\mathbf{L}} = 0)$ is the value of the objective function corresponding to the basis $\mathbf{B}$.

In Section C.2 we observed that linear programs always have extreme-point solutions and in this section we have focused on basic solutions. Are these concepts related? Indeed, they are. To conclude this section we show that extreme points and basic solutions are geometric and algebraic manifestations of the same concept.

Therefore, by finding an optimal basic solution, we are obtaining an optimal extreme point solution, and vice versa. Recall that columns of the constraint matrix $\mathcal{A}$ corresponding to the basis $\mathbf{B}$ are linearly independent.

Theorem C.1 (Extreme Points and Basic Feasible Solutions). *A feasible solution x to a linear program (C.1) is an extreme point if and only if the columns $\{\mathcal{A}_j : x_j > 0\}$ of the constraint matrix corresponding to the strictly positive variables are linearly independent.*

Proof. To simplify our notation, suppose that $x_1 > 0, x_2 > 0, \dots, x_k > 0$ and $x_{k+1} = x_{k+2} = \dots = x_q = 0$. Suppose that the columns $\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_k$ are linearly dependent. Then there exist constants $\omega_1, \omega_2, \dots, \omega_k$ (not all zero), satisfying the condition

$$\mathcal{A}_1\omega_1 + \mathcal{A}_2\omega_2 + \dots + \mathcal{A}_k\omega_k = 0.$$

Let us define $\omega_{k+1} = \omega_{k+2} = \dots = \omega_q = 0$ and let ω denote the vector $(\omega_1, \omega_2, \dots, \omega_q)$. Then since each of the first k components of the solution x are positive, for some sufficiently small value of the scalar θ ,

$$x + \theta\omega \geq 0 \quad \text{and} \quad x - \theta\omega \geq 0.$$

Also, since $\mathcal{A}\omega = 0$, $\mathcal{A}(x + \theta\omega) = \mathcal{A}(x - \theta\omega) = \mathcal{A}x = b$. Therefore, both $x + \theta\omega$ and $x - \theta\omega$ are feasible for the linear program. But then $x = \frac{1}{2}(x + \theta\omega) + \frac{1}{2}(x - \theta\omega)$, which implies that x is not an extreme point (because it lies on the line joining the points $x + \theta\omega$ and $x - \theta\omega$). Therefore, we have shown that if x is an extreme point, the columns $\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_k$ must be linearly independent.

To establish the converse, suppose that $x = \theta x^1 + (1 - \theta)x^2$ for some scalar $0 < \theta < 1$ and two feasible points x^1 and x^2 to the linear program (C.1). Since $x_j = \theta x_j^1 + (1 - \theta)x_j^2$ for any $j \geq k + 1$, and both x_j^1 and x_j^2 are nonnegative, both x_j^1 and x_j^2 must have value zero. Therefore, since both these points are feasible solutions to the linear program, they satisfy

$$\mathcal{A}_1 x_1^1 + \mathcal{A}_2 x_2^1 + \dots + \mathcal{A}_k x_k^1 = b,$$

and

$$\mathcal{A}_1 x_1^2 + \mathcal{A}_2 x_2^2 + \dots + \mathcal{A}_k x_k^2 = b.$$

Subtracting these equations from each other shows that

$$\mathcal{A}_1(x_1^1 - x_1^2) + \mathcal{A}_2(x_2^1 - x_2^2) + \dots + \mathcal{A}_k(x_k^1 - x_k^2) = 0.$$

But since the columns $\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_k$ are linearly independent, each component of x^1 and x^2 must be the same. But then we cannot represent $x = \theta x^1 + (1 - \theta)x^2$ in terms of two distinct feasible points of the linear program and therefore x is an extreme point. ♦

Although we have not stated this theorem as “ x is a basic feasible solution if and only if it is an extreme point,” the theorem easily implies this statement. Notice that if $k = p$ in this proof, the columns $\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_k$ form a basis of the linear program, so x is a basic feasible solution. On the other hand, if $k < p$, we can add

$p - k$ other linearly independent columns to $\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_k$ to form a basis $\mathcal{B}$. In this case x still is a basic feasible solution, but with some basic variables at value zero (those corresponding to the columns we have added). Therefore, these observations and Theorem C.1 imply that basic feasible solutions and extreme points are identical.

C.4 SIMPLEX METHOD

The simplex method maintains a basic feasible solution at every step. Given a basic feasible solution, the method first applies the optimality criteria to test the optimality of the current solution. If the current solution does not fulfill this condition, the algorithm performs an operation, known as a *pivot operation*, to obtain another basis structure with a lower or identical cost. The simplex method repeats this process until the current basic feasible solution satisfies the optimality criteria. Before describing the details of these steps, we consider some preliminary issues.

Obtaining an Initial Basis Structure

Recall from our preceding discussion that a basic solution might not be feasible. Therefore, identifying a basic feasible solution requires specifying a set of basic variables so that the columns associated with these variables are linearly independent and the solution x_B, x_L obtained by setting $x_B = \mathcal{B}^{-1}b$ and $x_L = 0$ is nonnegative. Unfortunately, there is no simple method for identifying any such collection of basic variables. In fact, finding a feasible solution of a linear program (or a basic feasible solution) is almost as difficult as finding an optimal solution. Nevertheless, using a simple technique, we can find a basic feasible solution of a related linear program and use it to initiate the simplex method. This technique consists of introducing an *artificial variable* x_{q+i} with a sufficiently large cost M for each constraint i and defining an augmented linear program as:

$$\text{Minimize } \sum_{j=1}^q c_j x_j + \sum_{j=q+1}^{q+p} M x_j \quad (\text{C.11a})$$

subject to

$$\sum_{j=1}^q a_{ij} x_j + x_{q+i} = b(i) \quad \text{for all } i = 1, \dots, p, \quad (\text{C.11b})$$

$$x_j \geq 0 \quad \text{for all } j = 1, \dots, q. \quad (\text{C.11c})$$

It is easy to show that the original linear program (C.1) has a feasible solution if and only if each artificial variable has value zero in every optimal solution of the augmented linear program (C.11). (The large cost coefficient associated with each artificial variable ensures that it has value zero in an optimal solution if the original problem has a feasible solution.) As a consequence, solving the augmented linear program solves the original linear program. For the augmented linear program, the set of all artificial variables constitutes an initial basis and the value of the i th basic variable is $b(i) \geq 0$.

Optimality Criteria

Let (B, L) denote a feasible basis structure of the linear program. Assume, for simplicity, that $B = \{1, 2, \dots, p\}$. Consider the canonical form associated with this basis structure. In this canonical form the objective function is

$$\text{Minimize } z(x) = z_0 + \sum_{j=p+1}^q c_j^{\pi} x_j. \quad (\text{C.12})$$

The coefficient $c_j^{\pi} = c_j - \sum_{i=1}^p \pi(i) a_{ij}$ is the reduced cost of the nonbasic variable x_j with respect to the current simplex multipliers π . We claim that if $c_j^{\pi} \geq 0$ for each nonbasic variable x_j , the current basic feasible solution x is an optimal solution of the linear program. To see this, observe that in any feasible solution of the linear program, $x_j \geq 0$ for all $j \in L$. Therefore, if $c_j^{\pi} \geq 0$ for each nonbasic variable x_j , then z_0 is a lower bound on the optimal objective value. As we noted in the preceding section, the current solution x , which sets $x_j = 0$ for all $j \in L$, achieves this lower bound and, therefore it must be optimal.

Pivot Operation

If $c_j^{\pi} < 0$ for some nonbasic variable x_j , the current basic feasible solution might not be optimal. The expression (C.12) implies that c_j^{π} is the rate of decrease in the objective function value per unit increase in the value of x_j . The simplex method selects one such nonbasic variable, say x_s , as the *entering variable* and tries to increase its value. As we will see, when the simplex method increases x_s as much as possible while keeping all the other nonbasic variable at value zero, some basic variable, say x_r , reaches value zero. The simplex method replaces the basic variable x_r by x_s , defining a new basis structure. It then updates the inverse of the basis and repeats the computations.

If we increase the value of the entering variable x_s to value θ and keep all other nonbasic variables at zero value, expression (C.8) implies that the basic variables x_B change in the following manner:

$$x_B + \theta \bar{\mathcal{A}}_s = \bar{b}. \quad (\text{C.13})$$

In this expression $\bar{b} = \mathcal{B}^{-1}b \geq 0$, $\bar{\mathcal{A}}_s = \mathcal{B}^{-1}\mathcal{A}_s$, and $x_B = [x_1, x_2, \dots, x_p]$. Let $\bar{\mathcal{A}}_s = [\bar{a}_{1s}, \bar{a}_{2s}, \dots, \bar{a}_{ps}]$. We can restate (C.13) as

$$x_i = \bar{b}(i) - \theta \bar{a}_{is} \quad \text{for all } i = 1, \dots, p. \quad (\text{C.14})$$

If $\bar{a}_{is} \leq 0$ and we increase θ , then x_i either remains unchanged or increases. If $\bar{a}_{is} > 0$ and we increase θ , then x_i decreases and eventually it will become zero. Consequently, (C.14) implies that if $\bar{a}_{is} > 0$, then the scalar θ must satisfy the condition $\theta \leq \bar{b}(i)/\bar{a}_{is}$ in order for x_i to remain nonnegative. As a result,

$$\theta = \min_{1 \leq i \leq p} \{ \bar{b}(i)/\bar{a}_{is} : \bar{a}_{is} > 0 \},$$

is the largest value of θ that we can assign to x_s while remaining feasible. What if $\bar{a}_{is} \leq 0$ for each $i = 1, \dots, p$? Then we can assign an arbitrarily large value to the entering variable x_s and the solution remains feasible. Since $c_s^{\pi} < 0$, by setting x_s as large as we like, we can make the objective function arbitrarily small, and make

it approach $-\infty$. In this instance we say that the linear program has an *unbounded solution*.

We next focus on situations in which θ is finite. If we set $x_s = \theta$, one of the basic variables, say x_r , becomes zero. Note that $\bar{b}(r)/\bar{a}_{rs} = \theta$. We refer to x_r as the *leaving variable* and refer to the rule we have described for identifying θ and the corresponding leaving variable as the *minimum ratio rule*. Next, we designate x_s a basic variable, x_r a nonbasic variable, and update the canonical form of the linear program so that it satisfies the canonical property with respect to the new basis. We do so by performing a sequence of elementary row operations.

Updating the Simplex Tableau

Recall from Section C.3 that in the canonical form with respect to the basis $\mathbf{B}$, the equations of the linear program assume the form

$$\mathbf{x}_B + \mathbf{B}^{-1}\mathcal{L}\mathbf{x}_L = \mathbf{B}^{-1}\mathbf{b}. \quad (\text{C.15})$$

In the new basis, the entering variable x_s becomes the basic variable for the r th row, which requires that in the new canonical form the variable x_s should have a coefficient $+1$ in the r th row, and a coefficient 0 in all other rows. We achieve this new canonical form by first dividing the r th row by $\bar{a}_{rs}$; the variable x_s then has a $+1$ coefficient in this row. Then, for each $1 \leq i \leq p$, $i \neq r$, we multiply the r th row by the constant $-\bar{a}_{is}$ and add it to the i th row so that the updated value of $\bar{a}_{is}$ becomes zero. We also multiply the r th row by a constant $-c_s^*$ and add it to the objective function so that the objective function coefficient of x_s becomes zero. We refer to this set of computations as a *pivot operation*.

To illustrate these steps of the simplex method, consider the linear programming example given in (C.6). For convenience, we state the example again.

$$\text{Minimize } z(x) = 0x_1 + 0x_2 + 3x_3 - x_4 + 10 \quad (\text{C.16a})$$

subject to

$$x_1 - 3x_3 + 3x_4 = 6, \quad (\text{C.16b})$$

$$x_2 - 8x_3 + 4x_4 = 4, \quad (\text{C.16c})$$

$$x_1, x_2, x_3, x_4 \geq 0. \quad (\text{C.16d})$$

The nonbasic variable x_4 has a negative reduced cost, and we select it as the entering variable. Applying the minimum ratio rule, we find that $\theta = \min\{6/3, 4/4\} = 1$, the minimum being achieved in the second row, which contains the basic variable x_2 . The variable x_2 is the leaving variable. In the next basis, x_1 and x_4 are the basic variables, so we need to modify the canonical form. Since x_4 is the new basic variable for the constraint (C.16c), we divide this constraint by 4 so that x_4 has a $+1$ coefficient. We then multiply the modified constraint by -3 and add it to (C.16b), and multiply it by $+1$ and add it to (C.16a). These operations produce the following (equivalent) formulation of the linear program:

$$\text{Minimize } z(x) = 0x_1 + 0x_4 + \frac{1}{4}x_2 + x_3 + 9$$

subject to

$$x_1 - \frac{3}{4}x_2 + 3x_3 = 3,$$

$$x_4 + \frac{1}{4}x_2 - 2x_3 = 1,$$

$$x_1, x_2, x_3, x_4 \geq 0.$$

In this canonical form, the reduced costs of both nonbasic variables x_2 and x_3 are nonnegative, so the current basic feasible solution, $x_1 = 3$, $x_4 = 1$, and $x_2 = x_3 = 0$, is optimal. This solution has an objective function value of 9 .

One way to perform the pivot operation is by updating the full matrix $\bar{\mathcal{A}}$, that is, by performing the explicit set of pivot computations iteratively on the matrix $\bar{\mathcal{A}}$. This set of computations can be very expensive for linear programs that contain many variables (as is typical in practice). The *revised simplex method* is a particular implementation of the simplex method that permits us to avoid many of these computations. To describe the basic approach of the revised simplex method, suppose that we (conceptually) append a set of fictitious variables y to the original linear program and form a new linear program with the constraints:

$$\bar{\mathcal{A}}x + \mathcal{I}y = b,$$

$$x \geq 0, \quad y \geq 0.$$

In this formulation, $\mathcal{I}$ is an identity matrix; that is, its diagonal elements all have value 1 and its elements off the diagonal all have value zero. Then to obtain the canonical form (C.13) with respect to the basis $\mathbf{B}$, we premultiply this system by the basis inverse $\mathbf{B}^{-1}$. With the fictitious variables y , the system becomes

$$\mathbf{x}_B + \mathbf{B}^{-1}\mathcal{L}\mathbf{x}_L + \mathbf{B}^{-1}y = \mathbf{B}^{-1}b.$$

As shown by this expression, if we were to perform the pivot operation on the entire matrix $\bar{\mathcal{A}}$ from step to step, the coefficients of fictitious variables y would be the basis inverse. This observation shows that we need not carry out the pivot operations on the entire matrix $\bar{\mathcal{A}}$. Instead, we can perform these operations on the columns associated with the initial identity matrix $\mathcal{I}$ (we do not formally introduce the variables y). Since the resulting computations give us the basis inverse $\mathbf{B}^{-1}$, we can use this matrix to compute the simplex multipliers $\pi = c_B\mathbf{B}^{-1}$ and then use them to compute the reduced cost of each variable. Once we have determined the variable x_s to introduce into the basis, we compute its representation $\bar{\mathcal{A}}_s = \mathbf{B}^{-1}\mathcal{A}_s$ and then use this information to perform the ratio test to identify the variable x_r to leave the basis. We next perform the row elementary operations on the current $\mathbf{B}^{-1}$ matrix and obtain the updated basis inverse.

The advantage of this approach is that we use only the original data $\bar{\mathcal{A}}$ in computing the reduced costs [using the formula $c_j^* = c_j - \sum_{i=1}^p \pi(i)a_{ij}$] and determine the modified data $\bar{\mathcal{A}}$ for only one column s of $\bar{\mathcal{A}}$ and perform row elementary operations only on the basis inverse matrix. This apparently modest change in the algorithm often has dramatic effects on its efficiency because the original data $\bar{\mathcal{A}}$ for most problems met in practice is very sparse in the sense that most (90 percent or more) of its coefficients are zero. On the other hand, the modified matrix $\bar{\mathcal{A}}$ typically becomes very dense as we perform the iterations of the simplex algorithm. In the revised simplex method, by using appropriate data structures, we can avoid all the

computations corresponding to the zero elements. As a consequence, by implementing the revised simplex method, we usually achieve great economies in our computations. The vast linear programming literature and many standard texts specify more details about this approach and about other ways to improve the empirical efficiency of the simplex method.

Finite Termination

The simplex method moves from one basic feasible solution to another by performing a pivot operation. In each iteration, the solution value improves by the amount $c_j^* \theta$. If $\theta > 0$, we say that the pivot is *nondegenerate*; otherwise, we refer to it as *degenerate*. A nondegenerate pivot strictly decreases the value of the objective function associated with the basic feasible solution. If every pivot is nondegenerate, the simplex method will terminate finitely because a linear program has at most qC_p distinct basic feasible solutions and the algorithm can never repeat any basic feasible solution (since the objective function value is strictly decreasing). However, the simplex method might perform degenerate pivots, and without further modifications it might repeat a basic feasible solution and therefore not terminate finitely. Fortunately, researchers have developed several ways for implementing the simplex method so that it converges finitely. We refer the reader to linear programming textbooks for a discussion of these techniques.

The simplex method terminates with one of the following three outcomes:

Case 1. The method terminates with an unbounded solution. In this case the linear program has no optimal solution with a finite solution value.

Case 2. The method terminates with a finite solution in which some artificial variable has a positive value. In this case the original linear program has no feasible solution.

Case 3. The method terminates with a finite solution in which all the artificial variables have zero value. This solution is an optimal solution of the original linear program.

C.5 BOUNDED VARIABLE SIMPLEX METHOD

Often linear programming problems have upper as well as lower bounds imposed on the variables. Network flow problems with arc capacities on the arc flows belong to this class. We refer to this class of linear programs, formulated as follows, as bounded variable linear programs.

$$\text{Minimize } \sum_{j=1}^q c_j x_j \quad (C.17a)$$

subject to

$$\sum_{j=1}^q a_{ij} x_j = b(i) \quad \text{for all } i = 1, \dots, p, \quad (C.17b)$$

$$x_j \leq u_j \quad \text{for all } j = 1, \dots, q, \quad (C.17c)$$

$$x_j \geq 0 \quad \text{for all } j = 1, \dots, q. \quad (C.17d)$$

The simplest way to handle the upper bound constraints (C.17c) is to treat them like the other constraints (by adding slack variables to convert them into equality constraints). However, doing so increases the size of the linear program substantially and is therefore undesirable. In this section we describe a generalization of the simplex method, called the *bounded variable simplex method*, that treats these upper bound constraints implicitly, very much like the lower bound constraints (C.17d).

In our previous discussion we defined a basic feasible solution in the simplex method by a pair $(\mathbf{B}, \mathbf{L})$, consisting of a set $\mathbf{B}$ of basic variables and a set $\mathbf{L}$ of nonbasic variables at their lower bounds. We define a basic feasible solution in the bounded variable simplex method by a triplet $(\mathbf{B}, \mathbf{L}, \mathbf{U})$, consisting of a set $\mathbf{B}$ of basic variables, a set $\mathbf{L}$ of nonbasic variables at their lower bounds, and a set $\mathbf{U}$ of nonbasic variables at their upper bounds. Let $\mathcal{B}$, $\mathcal{L}$, and $\mathcal{U}$ denote a compatible partitioning of the constraint matrix $\mathbf{A}$ corresponding to the sets $\mathbf{B}$, $\mathbf{L}$, and $\mathbf{U}$. Then

$$\mathcal{B}x_{\mathbf{B}} + \mathcal{L}x_{\mathbf{L}} + \mathcal{U}x_{\mathbf{U}} = b.$$

In a basic feasible solution we set the value of each nonbasic variable to its appropriate bound, depending on whether the variable is contained in $\mathbf{L}$ or $\mathbf{U}$. Let $u_{\mathbf{U}}$ denote the vector of upper bounds for variables in $\mathbf{U}$. Then, setting $x_{\mathbf{L}} = 0$ and $x_{\mathbf{U}} = u_{\mathbf{U}}$, we find that

$$\mathcal{B}x_{\mathbf{B}} = b - \mathcal{U}u_{\mathbf{U}},$$

or

$$x_{\mathbf{B}} = \mathcal{B}^{-1}b - \mathcal{B}^{-1}[\mathcal{U}u_{\mathbf{U}}].$$

As before, $\mathcal{B}^{-1}$ is the inverse of the basis $\mathcal{B}$. For the bounded variable simplex method, we define the simplex multipliers π in the same way as in the simplex method (i.e., as the multipliers to apply to the constraints to create the reduced cost of zero for each basic variable). We now discuss, one by one, various steps of the simplex method and point out any changes required in the bounded variable simplex method.

Optimality Criteria

In a canonical form of the linear program with respect to the basis $\mathbf{B}$, we can write the objective function as

$$z(x) = \sum_{j \in \mathbf{L}} c_j^* x_j + \sum_{j \in \mathbf{U}} c_j^* x_j + z_0.$$

Using arguments similar to those used earlier for the case without upper bounds, we can easily show that if $c_j^* \geq 0$ for each $j \in \mathbf{L}$ and $c_j^* \leq 0$ for each $j \in \mathbf{U}$, then $z_0 + \sum_{j \in \mathbf{U}} c_j^* u_j$ is a lower bound on the optimal value of the objective function. We refer to these conditions as the *optimality* conditions for bounded variable linear programs. If the basic feasible solution satisfies these optimality conditions, the objective function achieves this lower bound, and hence it must be optimal. Observe that the optimality conditions imply that it is not cost-effective to

increase the value of any nonbasic variable at its lower bound or decrease the value of any nonbasic variable at its upper bound.

Entering Arc Criteria

The bounded variable simplex method selects any nonbasic variable violating its optimality condition as the entering arc. In other words, a nonbasic variable x_j is eligible to be an entering arc if (1) $j \in L$ and $c_j^* < 0$, or (2) $j \in U$ and $c_j^* > 0$.

Leaving Arc Criteria

If the entering variable x_s is a nonbasic variable at its lower bound, the method attempts to increase the value of x_s by the largest possible amount, and if the entering variable x_s is at its upper bound, the method attempts to decrease its value by the largest possible amount. The maximum change is determined by the requirement that the value of every basic variable and of the nonbasic variable x_s remains between (or at) its lower and upper bounds. The expression (C.14) implies that if $\bar{a}_{is} < 0$, then by increasing θ , we increase the value of the basic variable x_i . In this case the maximum change allowed by the upper bound constraint of x_i is $\theta_i = (u_i - \bar{b}(i))/(-\bar{a}_{is})$. If $\bar{a}_{is} > 0$, then by increasing θ , we decrease the value of the basic variable x_i . In this case the maximum change allowed by the lower bound constraint of x_i is $\theta_i = \bar{b}(i)/\bar{a}_{is}$. We let $\theta_i = +\infty$ if $\bar{a}_{is} = 0$. Finally, the maximum change allowed by the upper and lower bound constraints of x_s is $|u_s|$. The maximum value of θ that we can assign to x_s will be the minimum of $|u_s|$ and $\min\{\theta_i : 1 \leq i \leq p\}$. In the former case the basis remains unchanged; x_s simply moves from the set L to U or from the set U to L . In the latter case the basis changes and so does the canonical form. The method of updating the canonical form is same as for the case without upper bounds. Conceivably, we might find that $\theta = \infty$, in which case the linear programming problem has an unbounded solution. The case, when the entering variable x_s is a nonbasic variable at its upper bound, is left as an exercise to the reader.

Termination of the Method

Under the nondegeneracy assumption, the bounded variable simplex method terminates finitely because it strictly decreases the objective function value of the basic feasible solution at each step and any linear program has a finite number of basic feasible solutions. Without the nondegeneracy assumption, we need some additional technique to ensure finite termination of the method. For a discussion of these techniques, we refer the reader to linear programming textbooks.

C.6 LINEAR PROGRAMMING DUALITY

Each linear programming problem, which for the purposes of this discussion we call the *primal problem*, has a closely related associated linear programming problem, called the *dual problem*, and together these two problems define a duality theory that lies at the heart of linear programming and many other areas in the field of constrained optimization. In this section we review briefly some of the most salient features of duality theory.

While discussing duality theory, we assume that the linear program has been stated in the following form:

$$\text{Minimize } \sum_{j=1}^q c_j x_j \quad (\text{C.18a})$$

subject to

$$\sum_{j=1}^q a_{ij} x_j \geq b(i) \quad \text{for all } i = 1, \dots, p, \quad (\text{C.18b})$$

$$x_j \geq 0 \quad \text{for all } j = 1, \dots, q. \quad (\text{C.18c})$$

In this formulation, we permit $b(i)$ to have an arbitrary sign. We refer to this form as the *symmetric form* of a linear program. It is possible to show that we can convert any linear program in a nonsymmetric form into the symmetric form. To define the dual problem, we associate a dual variable $\pi(i)$ with the i th constraint in (C.18b). With respect to these variables, the dual problem is:

$$\text{Maximize } \sum_{i=1}^p b(i)\pi(i) \quad (\text{C.19a})$$

subject to

$$\sum_{i=1}^p a_{ij}\pi(i) \leq c_j \quad \text{for all } j = 1, \dots, q, \quad (\text{C.19b})$$

$$\pi(i) \geq 0 \quad \text{for all } i = 1, \dots, p. \quad (\text{C.19c})$$

Notice that each constraint in the primal has an associated variable in the dual and that the right-hand side of this constraint becomes the cost coefficient of the associated variable. Moreover, each variable in the primal has an associated constraint in the dual and the cost coefficient of this variable becomes the right-hand side of the associated constraint. It is easy to verify that the dual of the dual is the primal.

In the linear programming problem, whose dual we want to form, if some constraint $\sum_{j=1}^q a_{ij} x_j = b(i)$ is in the equality form, we form the dual exactly as described earlier, except that the dual variable $\pi(i)$ becomes unrestricted in sign. Similarly, in the primal, if some primal variable x_i is unrestricted in sign, then in the dual the constraint corresponding to x_i is an equality constraint. Our first result concerning the primal-dual pair is known as the *weak duality theorem*.

Theorem C.2 (Weak Duality Theorem). If x is any feasible solution of the primal problem and π is any feasible solution of the dual problem, then $\sum_{j=1}^q c_j x_j \leq \sum_{i=1}^p b(i)\pi(i)$.

Proof. Multiplying the i th constraint in (C.18b) by $\pi(i)$ and adding yields

$$\sum_{i=1}^p b(i)\pi(i) \leq \sum_{i=1}^p \pi(i) \left[\sum_{j=1}^q a_{ij} x_j \right], \quad (\text{C.20})$$

while multiplying the j th constraint in (C.19b) by x_j and adding yields

$$\sum_{j=1}^q x_j \left[\sum_{i=1}^p a_{ij} \pi(i) \right] \leq \sum_{j=1}^q c_j x_j. \quad (\text{C.21})$$

Since the right-hand side of (C.20) equals the left-hand side of (C.21), together these two constraints imply that

$$\sum_{i=1}^p b(i) \pi(i) \leq \sum_{j=1}^q c_j x_j,$$

which is the conclusion of the lemma. $\blacklozenge$

The weak duality theorem has a number of immediate consequences, which we state next.

Property C.3

- (a) The objective function value of any feasible dual solution is a lower bound on the objective function value of every feasible primal solution.
- (b) If the primal problem has an unbounded solution, the dual problem is infeasible.
- (c) If the dual problem has an unbounded solution, the primal problem is infeasible.
- (d) If the primal problem has a feasible solution x and the dual problem has a feasible solution π and $\sum_{i=1}^p b(i) \pi(i) = \sum_{j=1}^q c_j x_j$, then x is an optimal solution of the primal problem and π is an optimal solution of the dual problem.

Theorem C.4 (Strong Duality Theorem). If any one of the pair of primal and dual problems has a finite optimal solution, so does the other one and both have the same objective function values.

Proof. Assume without any loss of generality that the primal problem has a finite optimal solution. We can convert the primal problem into standard form (in which all constraints have equalities instead of inequalities) by adding slack variables. Suppose that we apply the simplex method to this standard form and let $(\mathbf{B}, \mathbf{L})$ be the optimal basis structure. Let x be the optimal solution and π be the simplex multipliers associated with the optimal basis structure. Recall from our previous discussion that the simplex multipliers π are the multipliers associated with the constraints (C.1b), which when subtracted from the original form of the objective function yield the objective function in the final canonical form. Therefore,

$$z(x^*) = \sum_{j \in \mathbf{B}} c_j^\pi x_j + \sum_{j \in \mathbf{L}} c_j^\pi x_j + \sum_{i=1}^p b(i) \pi(i),$$

and

$$c_j^\pi = c_j - \sum_{i=1}^p a_{ij} \pi(i). \quad (\text{C.22})$$

Since x is a basic feasible solution, $c_j^\pi = 0$ for all $j \in \mathbf{B}$, and $x_j = 0$ for all $j \in \mathbf{L}$. Consequently, $z(x^*) = \sum_{i=1}^p b(i) \pi(i)$. Moreover, since the basis structure $(\mathbf{B}, \mathbf{L})$ satisfies the optimality criteria (i.e., $c_j^\pi \geq 0$ for all $j \in \mathbf{B} \cup \mathbf{L}$), the expressions (C.22) and (C.19b) imply that π is a feasible solution to the dual problem. The objective function value of this dual solution is $\sum_{i=1}^p b(i) \pi(i)$, which is same as that

of the primal solution x . Property C.3(d) implies that π is an optimal dual solution, completing the proof of the theorem. $\blacklozenge$

The weak and strong duality theorems imply several fundamental results concerning relationships between the primal and dual problems. The following complementary slackness property, which is another way of relating the two problems, makes some of these relationships more explicit.

Complementary Slackness Property. A pair (x, π) of the primal and dual feasible solutions is said to satisfy the complementary slackness property if

$$\pi(i) \left[\sum_{j=1}^q a_{ij} x_j - b(i) \right] = 0 \quad \text{for each } i = 1, \dots, p, \quad (\text{C.23a})$$

and

$$x_j \left[c_j - \sum_{i=1}^p a_{ij} \pi(i) \right] = 0 \quad \text{for each } j = 1, \dots, q. \quad (\text{C.23b})$$

Observe from the formulation (C.18) of the primal problem that $[\sum_{j=1}^q a_{ij} x_j - b(i)]$ is the amount of slack in the i th primal constraint and $\pi(i)$ is the dual variable associated with this constraint. Similarly, $[c_j - \sum_{i=1}^p a_{ij} \pi(i)]$ is the amount of slack in the j th dual constraint and x_j is the primal variable associated with this constraint. The complementary slackness property states that for every primal and dual constraint, the product of the slack in the constraint and its associated primal or dual variable is zero. In other words, if a constraint has a positive slack, the associated primal or dual variable must have value zero, and alternatively, if a primal or dual variable has a positive value, the dual solution must satisfy the corresponding constraint as an equality.

Theorem C.5 (Complementary Slackness Optimality Conditions). A primal feasible solution x and a dual feasible solution π are optimal solutions of the primal and dual problems if and only if they satisfy the complementary slackness property.

Proof. We first prove that if x and π are optimal primal and dual solutions, they must satisfy the complementary slackness property. While proving the weak duality theorem, we observed that

$$\sum_{i=1}^p b(i) \pi(i) \leq \sum_{i=1}^p \sum_{j=1}^q a_{ij} \pi(i) x_j \leq \sum_{j=1}^q c_j x_j. \quad (\text{C.24})$$

Since x and π are optimal primal and dual solutions, the strong duality theorem implies that both of the inequalities in (C.24) must be satisfied as equalities. We can rewrite the first equality in (C.23) as

$$\sum_{i=1}^p \pi(i) \left[\sum_{j=1}^q a_{ij} x_j - b(i) \right] = 0. \quad (\text{C.25})$$

Since x and π are feasible in the primal and dual problems, each term in (C.25)

is nonnegative. Consequently, the sum of these terms can be zero only if each term is individually zero. This observation shows that the solutions satisfy (C.23a). Similarly, the second equality in (C.24) implies that the solutions satisfy (C.23b).

We next prove the converse result: namely, if the solutions x and π satisfy the complementary slackness property, they must be optimal in the primal and dual problems. Adding (C.23a) for all i yields

$$\sum_{j=1}^q \sum_{i=1}^p a_{ij} \pi(i) x_j = \sum_{i=1}^p b(i) \pi(i).$$

Similarly, adding (C.23b) for all j yields

$$\sum_{j=1}^q \sum_{i=1}^p a_{ij} \pi(i) x_j = \sum_{j=1}^q c_j x_j.$$

Therefore, $\sum_{i=1}^p b(i) \pi(i) = \sum_{j=1}^q c_j x_j$, and by Property C.3(c), x is an optimal primal solution and π is an optimal dual solution, concluding the proof of the theorem. ♦

REFERENCE NOTES

Dantzig, who conducted the pioneering work in linear programming, developed the simplex method in 1947 to solve several military planning problems. The optimization community introduced a steady stream of developments since then and now the theory of linear programming includes a vast body of knowledge. Books by Dantzig [1962], Bradley, Hax, and Magnanti [1977], Chvátal [1983], Schrijver [1986], and Winston [1991] are excellent references on the history, applications, and theory of this topic.

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